

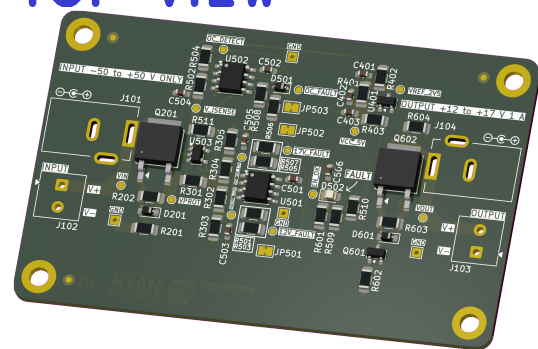
INPUT PROTECTION

VARIANT: DRAFT

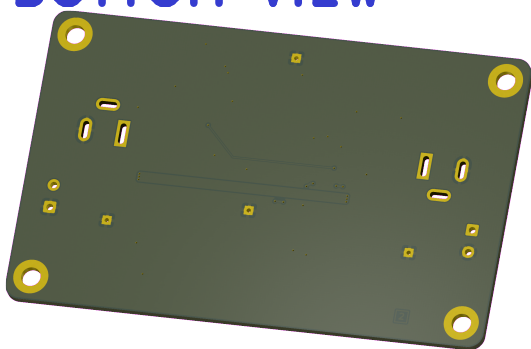
2026-02-18
REV + (Unreleased)

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TOP VIEW



BOTTOM VIEW



NOTES

COMMENT

NOT FITTED COMPONENTS MARKED AS **X**

DRAFT – EARLY STAGE SCHEMATIC CAPTURE
PRELIMINARY – LATE STAGE SCHEMATIC CAPTURE
CHECKED – PCB LAYOUT CHECKED, CONTACT ENGINEER IF MISTAKE FOUND
RELEASED – READY FOR PRODUCTION

DESIGN CONSIDERATIONS

DESIGN NOTE:

Example text for informational design notes.

DESIGN NOTE:

Example text for debug notes.

DESIGN NOTE:

Example text for cautionary design notes.

DESIGN NOTE:

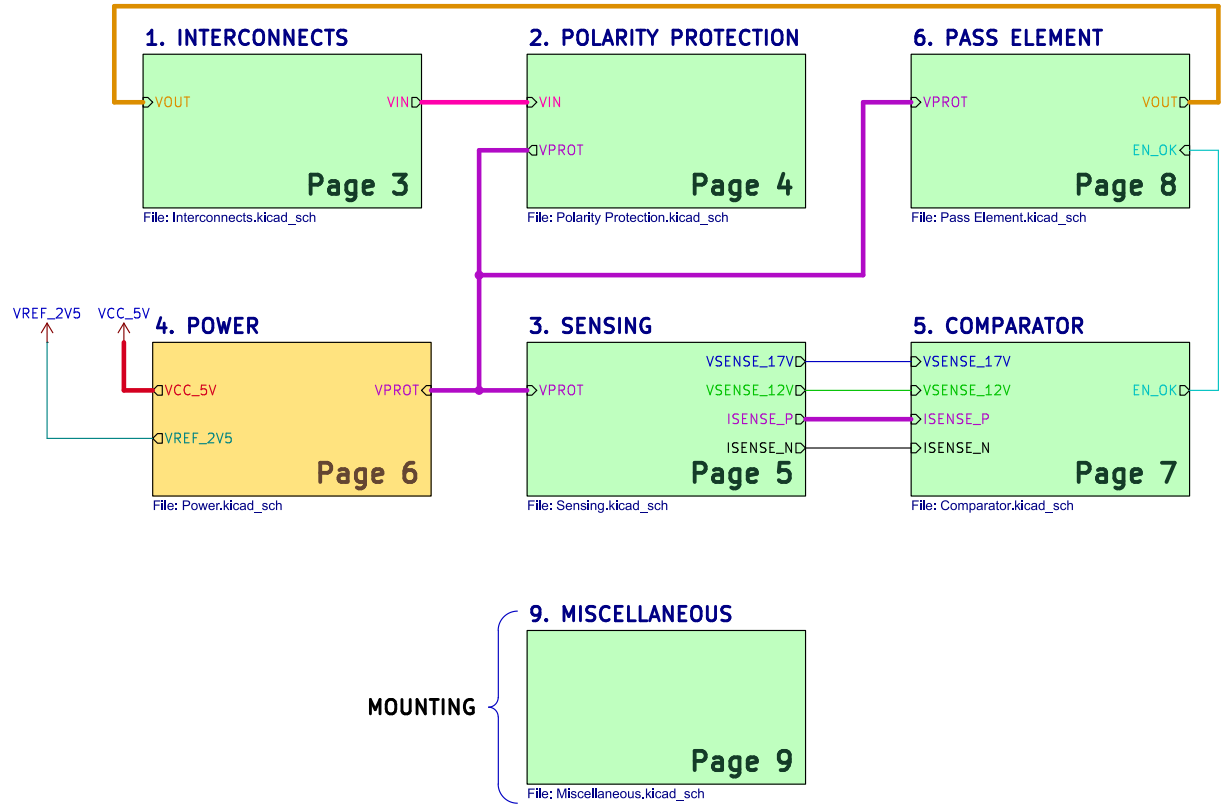
Example text for critical design notes.

LAYOUT NOTE:

Example text for critical layout guidelines.

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DRN			
2025-01-12 R. HICKS			
CHK	INPUT PROTECTION		
ENG APP			
MFR APP			
SHEET PATH	GIT HASH	DRAWING No	A3
/	1c5c101		
FILENAME	Input-Protection.kicad_sch	VARIANT	DRAFT
		REVISION + (Unreleased)	SHEET
			1 OF 10

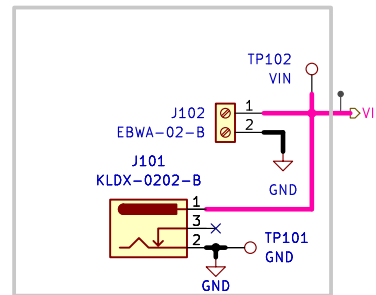
[2] ARCHITECTURE



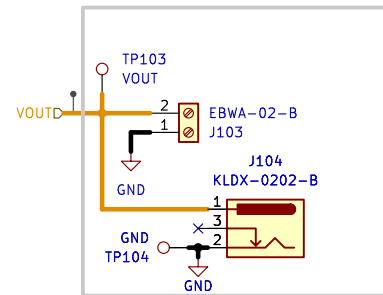
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DRN 2025-01-12 R. HICKS			
CHK			
ENG APP	INPUT PROTECTION		
MFR APP			
SHEET PATH /ARCHITECTURE/	GIT HASH 1c5c101	DRAWING No ARCHITECTURE	A3
FILENAME Project Architecture.kicad_sch	VARIANT DRAFT	REVISION + (Unreleased) SHEET 2 OF 10	

[3] 1. INTERCONNECTS

INPUT



OUTPUT



LAYOUT NOTE:
Large traces for VIN and VOUT, with 1oz copper a track width of >0.8mm is required to handle 2A of current with a FoS=2.

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/ARCHITECTURE/1. INTERCONNECTS/

FILENAME Interconnects.kicad_sch



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1. INTERCONNECTS

A4

VARIANT

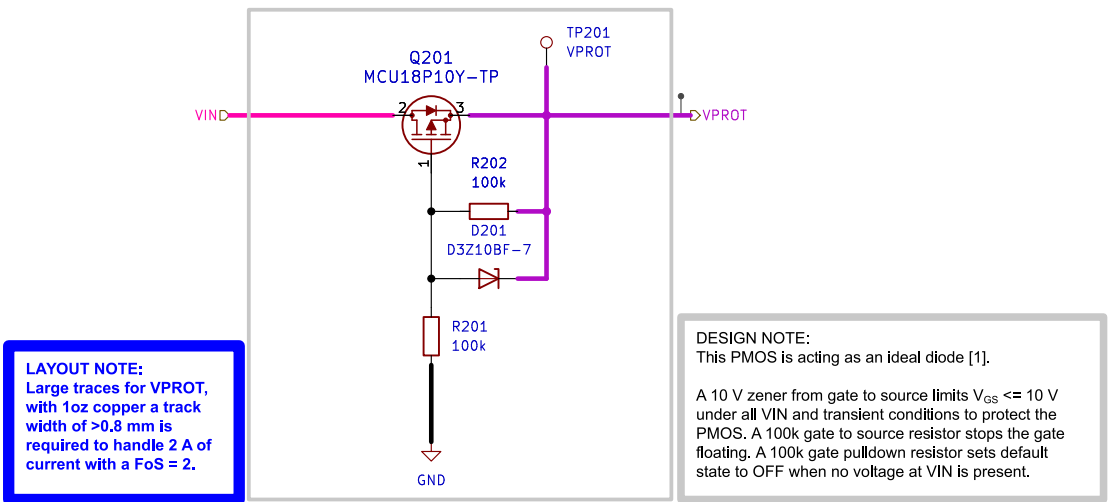
DRAFT

REVISION + (Unreleased)

SHEET

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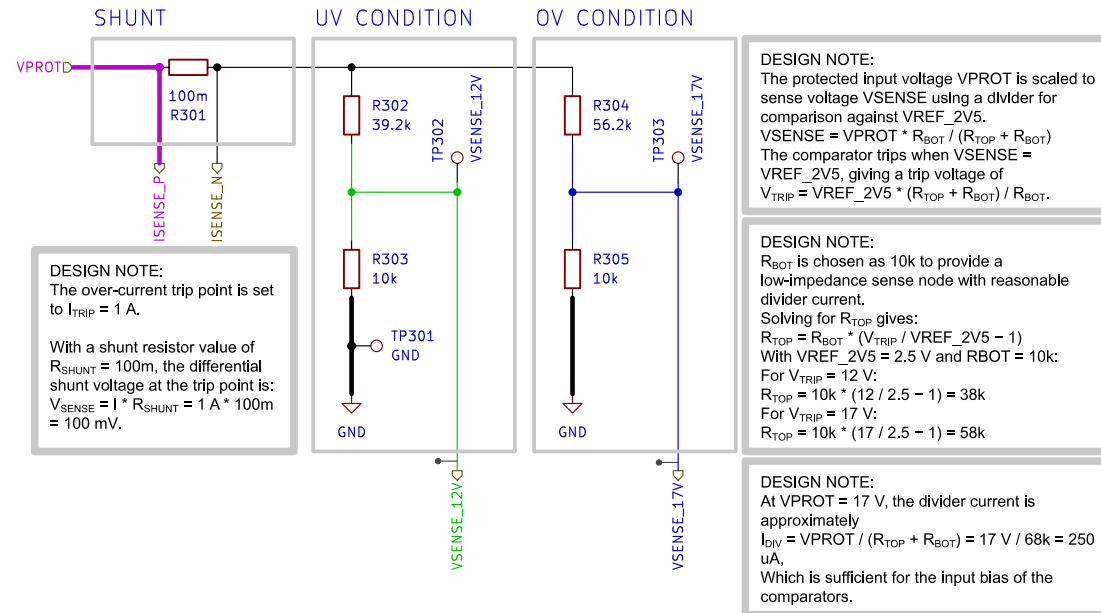
[4] 2. POLARITY PROTECTION



[1] <https://www.ti.com/lit/an/slvae57b/slvae57b.pdf?ts=1770319054914>

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CHK			
ENG APP		INPUT PROTECTION	
MFR APP			
SHEET PATH			
/ARCHITECTURE/2. POLARITY PROTECTION/		GIT HASH	DRAWING No
		1c52101	POLARITY PROTECTION
FILENAME	Polarity Protection.kicad_sch	VARIANT	DRAFT
		REVISION + (Unreleased)	SHEET
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[5] 3. SENSING



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SHEET PATH

/ARCHITECTURE/3. SENSING/

FILENAME Sensing.kicad_sch



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3. SENSING

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VCC_5V

The diagram illustrates the internal circuitry of the VREF and VCC_5V sections of the ADXL345 breakout board. It is divided into two main functional blocks: VREF_2V5 and VCC_5V.

VREF_2V5 Section: This section provides a precise 2.5V reference voltage. It features a voltage divider network consisting of two 100k resistors, R401 and R402. R401 is connected to the VREF_2V5 input, and R402 is connected to ground. A 10nF capacitor, C401, is connected in parallel with R402 to ground to filter noise. The output of the divider is labeled TP401 and VREF_2V5.

VCC_5V Section: This section provides a regulated 5V supply. It includes a 5V precision centrerail voltage detector, U401 (LM4040N-5.0), which is configured as a voltage detector. A 3.32k resistor, R403, is connected between the VCC_5V input and the detector's input. The detector's output is connected to the VCC_5V output. A 1uF capacitor, C402, is connected between the VCC_5V input and ground, and a 100nF capacitor, C403, is connected between the VCC_5V output and ground. The output is labeled TP402 and VCC_5V.

Other Components: A VPROT pin is shown at the top, which is a protection pin. The ground connections for both sections are clearly marked as GND.

[2] <https://www.ti.com/lit/ds/symlink/lm4040-n.pdf?ts=1752678691365>

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ENG APP				
MFR APP		INPUT PROTECTION		
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1c5c101		4. POWER		
FILENAME		REVISION + (Unreleased)		
Power.kicad_sch		DRAFT		6 OF 10

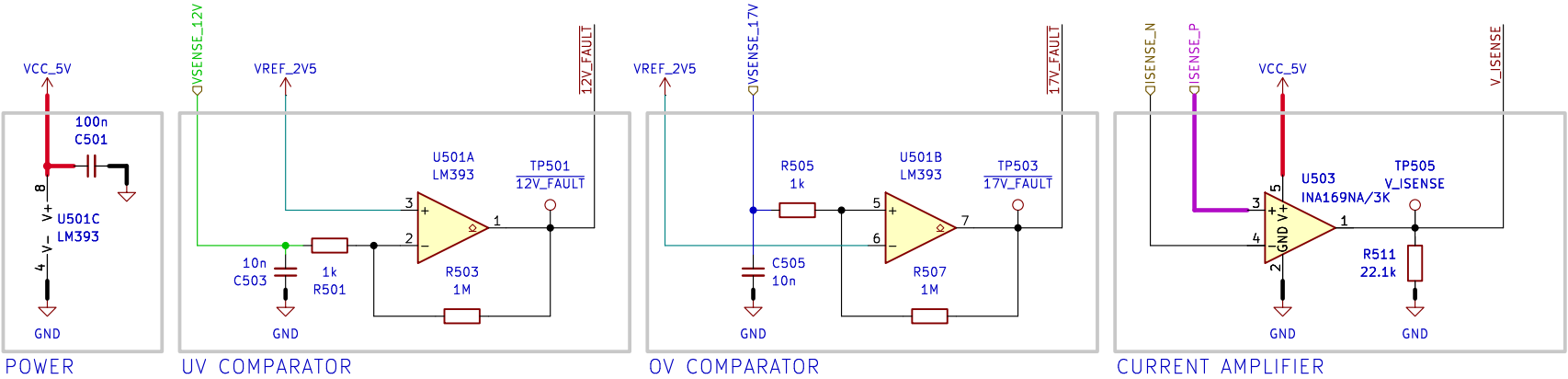
[7] 5. COMPARATOR

[3] <https://www.ti.com/lit/an/snoaa35f/snoaa35f.pdf?ts=1770697404856>

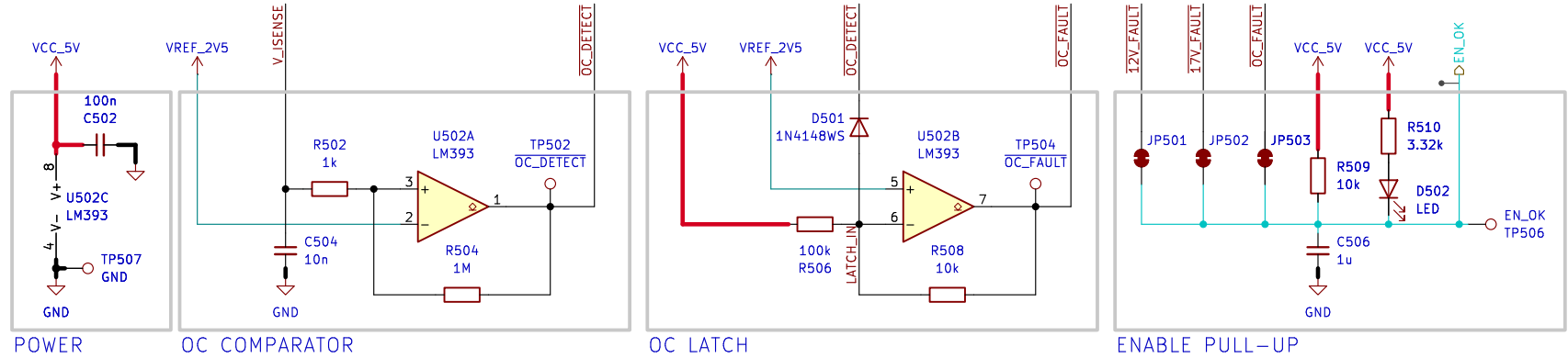
DESIGN NOTE:
LM393 dual comparators are used to implement under/over-voltage detection for the 12 V and 17 V supply thresholds. Each comparator compares a scaled VSENSE voltage against a 2.5 V precision reference (VREF_2V5) to generate active-low fault signals (12V_FAULT and 17V_FAULT). The LM393 is selected for its open-collector outputs.

DESIGN NOTE:
RC low-pass filters are placed on each VSENSE input to reduce noise and minimise chatter. The filters have a cutoff frequency of: $f_c = 1 / (2\pi * 1k * 10n) = 16 \text{ Hz}$. Positive feedback resistors provide hysteresis to again reduce chatter. Capacitor placement in accordance with 6.11 [3].

DESIGN NOTE:
Load current is measured using an INA169 current-sense amplifier. With a transconductance of 1 mA/V, the output current produces a voltage across the load, with $R_{SHUNT}=100m$ and $I_{TRIP}=1A$ with a 2.5 V reference we pick ~25k for R_L (R511). This corresponds to $V_{SENSE}=2.5 \text{ V}$ at 1 A.



LAYOUT NOTE:
Route ISENSE leads short from shunt to high-side current comparator.



DESIGN NOTE:
Solder bridges are provided to isolate different fault signals, cut the trace and re-solder to reconnect.

DESIGN NOTE:
A second LM393 comparator monitors the V_ISENSE signal from the current-sense amplifier and compares it against the 2.5 V reference to detect over-current conditions. When the sensed current exceeds the programmed threshold, the comparator output asserts an active-low over-current fault signal (OC_FAULT).

DESIGN NOTE:
The spare LM393 comparator is configured as a latch. This prevents "hiccups". OC_DETECT is reverse bias diode-coupled into the latch input. Once an over-current condition occurs, the latch forces OC_FAULT low and maintains the fault state until power is removed.

DESIGN NOTE:
All active-low fault signals (12V_FAULT, 17V_FAULT, and OC_FAULT) are combined using wired-AND logic enabled by the open-collector comparator outputs. A single pull-up resistor biases EN_OK high during normal operation. Assertion of any fault pulls EN_OK low, disabling the pass MOSFET.

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Comparator.kicad_sch



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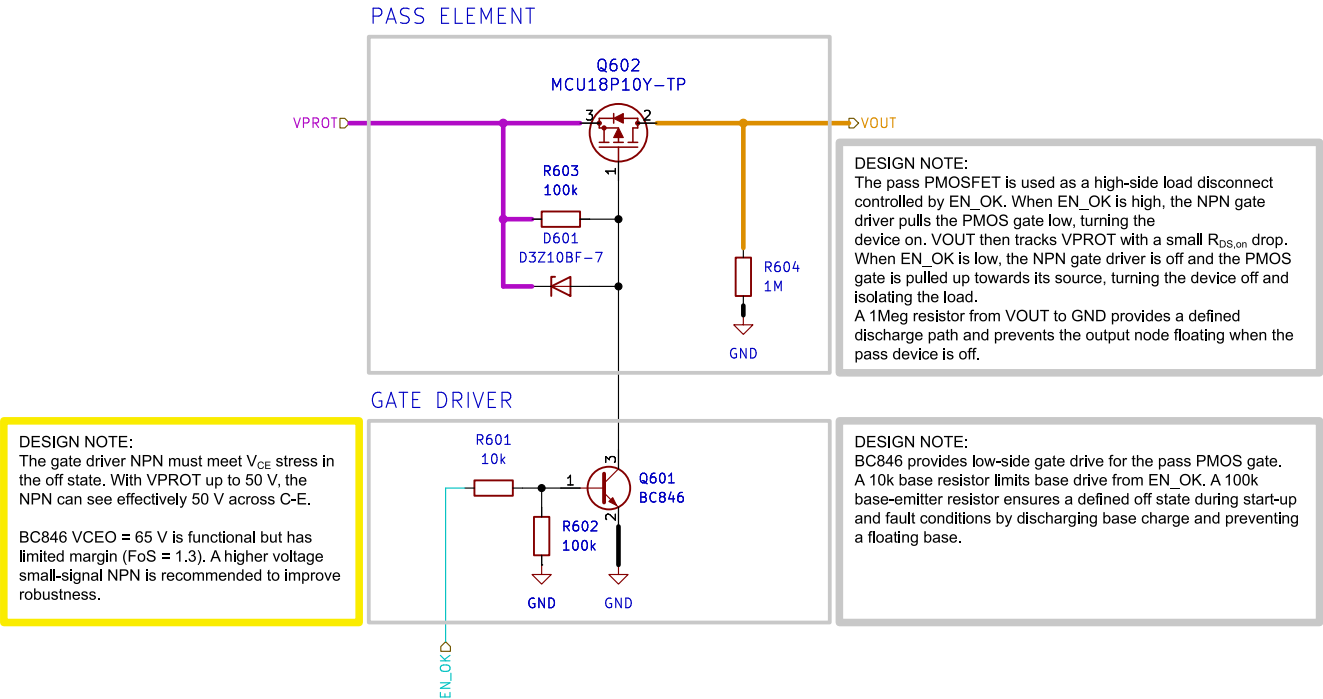
1c5c101

5. COMPARATOR

A3

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[8] 6. PASS ELEMENT



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ASSIGNMENT 2

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CHK

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MFR APP

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/ARCHITECTURE/6. PASS ELEMENT/

FILENAME Pass Element.kicad_sch



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6. PASS ELEMENT

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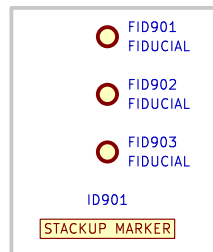
SHEET

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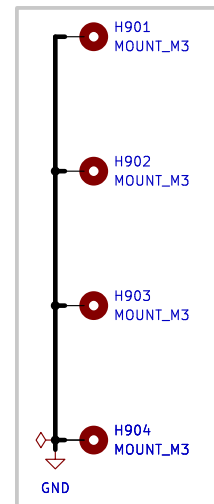
A4

[9] 9. MISCELLANEOUS

MARKINGS



MOUNTING



PROJECT
ASSIGNMENT 2

DRN
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/ARCHITECTURE/9. MISCELLANEOUS/

FILENAME Miscellaneous.kicad_sch



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VARIANT

DRAWING No

9. MISCELLANEOUS

REVISION + (Unreleased)

SHEET

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[10] REVISION HISTORY

Version 1.0.0 – 2026-02-11

Fixed

- Tidied up some routing issues

Added

- N/A

Changed

- Changed fiducials to be circular
- Changed info block

Removed

- N/A

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/REVISION HISTORY/

FILENAME Revision History.kicad_sch



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VARIANT

DRAFT

REVISION + (Unreleased)

SHEET

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